Functional Requirements Document (FRD)

Logistics Analytics Dashboard | Tableau Internship Project

1. Project Title

Logistics Analytics Dashboard

The project title is aligned with the BRD and covers the Tableau dashboard built for logistics revenue, delivery performance, cost, warehouse, and operational intelligence.

2. Dashboard Sections

Overview Section:

- Executive Overview page showing total revenue, total orders, on-time rate, average rating, order volume and revenue trend, orders and average rating by country, and revenue by category and region.

Drilldown Section(s):

- Delivery Performance page for average delivery days, late delivery rate, carrier performance, delivery day distribution, and region-quarter delay patterns.
- Revenue & Cost Analysis page for gross revenue, net revenue, discount impact, monthly revenue comparison, discount impact by category, and shipping cost by carrier over time.
- Operations & Warehouse Intelligence page for average weight per order, total quantity shipped, warehouse order volume, average weight by warehouse, driver workload, and weight distribution by category.

3. Data Requirements

Dashboard Section	Data Fields Needed	Source
Overview	Order_ID, Order_Date, Country, Region, Category, Quantity, Unit_Price, Discount, Delivery_Days, Customer_Rating	Logistic Clean.xlsx / Fact_Logistics and Date, Country, Region, Category dimensions
Delivery Performance	Order_ID, Carrier, Region, Order_Date, Delivery_Date, Delivery_Days, Customer_Rating, Status	Logistic Clean.xlsx / Fact_Logistics with Carrier, Region, Date, Status dimensions
Revenue & Cost Analysis	Order_ID, Order_Date, Category, Carrier, Quantity, Unit_Price, Discount, Shipping_Cost	Logistic Clean.xlsx / Fact_Logistics with Date, Category, Carrier dimensions

Operations & Warehouse Intelligence	Order_ID, Warehouse_ID, Driver_ID, Category, Weight_kg, Quantity, Carrier, Order_Date	Logistic Clean.xlsx / Fact_Logistics with Date, Category, Carrier dimensions

4. Filters / Slicers

- Year of Date / Date Range
- Carrier
- Category
- Region
- Country
- Status
- Priority
- Warehouse ID
- Delivery Speed Group

5. Visuals / Charts

Dashboard Section	Visual Type(s)
Executive Overview	KPI cards, dual-axis bar/line trend, filled map, treemap, year filter.
Delivery Performance	KPI cards, carrier scatter plot, region-quarter heatmap, delivery day histogram/distribution chart, carrier and year filters.
Revenue & Cost Analysis	KPI cards, monthly gross vs net revenue combo chart, packed bubble chart for discount impact, stacked area chart for shipping cost by carrier, category and year filters.
Operations & Warehouse Intelligence	KPI cards, warehouse bar and dot combo chart, horizontal driver workload bar chart, box-and-whisker plot for weight distribution, carrier/year/category filters.

6. Interactivity

- Filters should update all relevant charts within the selected dashboard page.
- Cross-filtering should allow users to click a region, category, country, or carrier and update related visual context.
- Navigation tabs should allow users to move between Executive Overview, Delivery Performance, Revenue & Cost Analysis, and Operations & Warehouse Intelligence.
- Tooltips should display supporting values such as order count, revenue, average rating, delivery days, discount impact, and shipping cost.
- Drilldown should support time-based analysis from year to quarter/month where applicable.
- Map and treemap interactions should help users identify country, region, and category-level performance quickly.

7. Calculations / Measures

Calculation Name	Formula	Purpose
Gross Revenue	SUM([Quantity] * [Unit_Price])	Calculates total revenue before discount.
Net Revenue	SUM(([Quantity] * [Unit_Price]) * (1 - [Discount]))	Calculates revenue after discount deduction.
Total Orders	COUNTD([Order_ID])	Counts unique orders processed.
Average Rating	AVG([Customer_Rating])	Measures customer satisfaction.
Average Delivery Days	AVG([Delivery_Days])	Measures average delivery duration.
Late Delivery Flag	IF [Delivery_Days] > 15 THEN 1 ELSE 0 END	Identifies delayed orders based on delivery-day threshold.
Late Delivery Rate %	SUM([Late Delivery Flag]) / COUNTD([Order_ID])	Calculates percentage of delayed orders.
On-Time Rate %	1 - [Late Delivery Rate %]	Calculates percentage of orders delivered on time.
Discount Impact Value	SUM([Quantity] * [Unit_Price] * [Discount])	Measures revenue reduced due to discount.
Discount Impact %	[Discount Impact Value] / [Gross Revenue]	Shows discount effect as a percentage of gross revenue.
Total Shipping Cost	SUM([Shipping_Cost])	Measures logistics cost by carrier/time.
Average Weight Per Order	AVG([Weight_kg])	Measures average shipment weight.
Total Quantity Shipped	SUM([Quantity])	Measures total units shipped.
Driver Workload	COUNTD([Order_ID]) by [Driver_ID]	Shows order load assigned to each driver.
Delivery Speed Group	IF [Delivery_Days] <= 7 THEN "Fast" ELSEIF [Delivery_Days] <= 14 THEN "Normal" ELSE "Slow" END	Groups deliveries into speed categories for distribution analysis.

8. Notes / Special Instructions

- Use the cleaned logistics dataset as the primary source for all dashboard visuals and measures.
- Maintain the star-schema model shown in the project screenshots: Fact_Logistics connected to Date, Carrier, Category, Region, Country, Status, Priority, and Customer dimensions.
- Use consistent KPI formatting: revenue in whole numbers or millions, percentages with two decimals, and average measures with two decimals where appropriate.
- Unknown values such as Unknown Carrier, Unknown Region, and WH-UNKNOWN should remain visible to avoid hiding data quality gaps.
- Dashboard design should prioritize readability, consistent color usage, clear titles, and business-friendly labels.
- Calculations should be validated against dashboard KPI cards before final publishing.
- Final Tableau workbook should be version controlled and published on Day 5 as per the internship project plan.